HERIOT WATT UNIVERSITY



B37VR ROBOTICS PROJECT

3 Characterising motors and driving robot

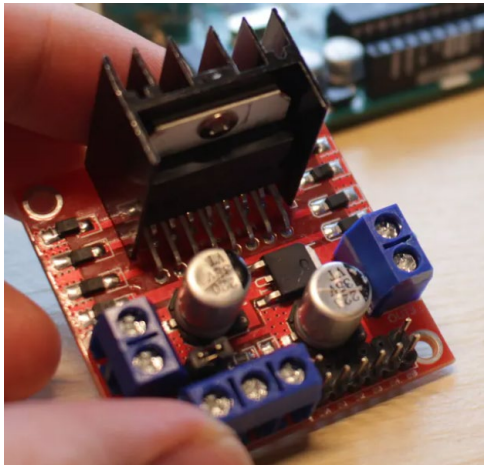
January '21

3 SUMMARY

- **LEARNING RESOURCES AVAILABLE**
  - Kit of parts to make robot
  - Motor drive code
- **LEARNING ACTIVITIES**
  - Show operation of motors
  - Discuss what is needed in characterisation experiments
- **TASKS TO BE COMPLETED**
  - Build up robot kit
  - Attach Arduino board, L298 board and battery to robot
  - Wire up motors to operate
  - Adjust speed of motor by editing code
  - Characterise the robot for driving motors at different speeds as indicated in separate sheet

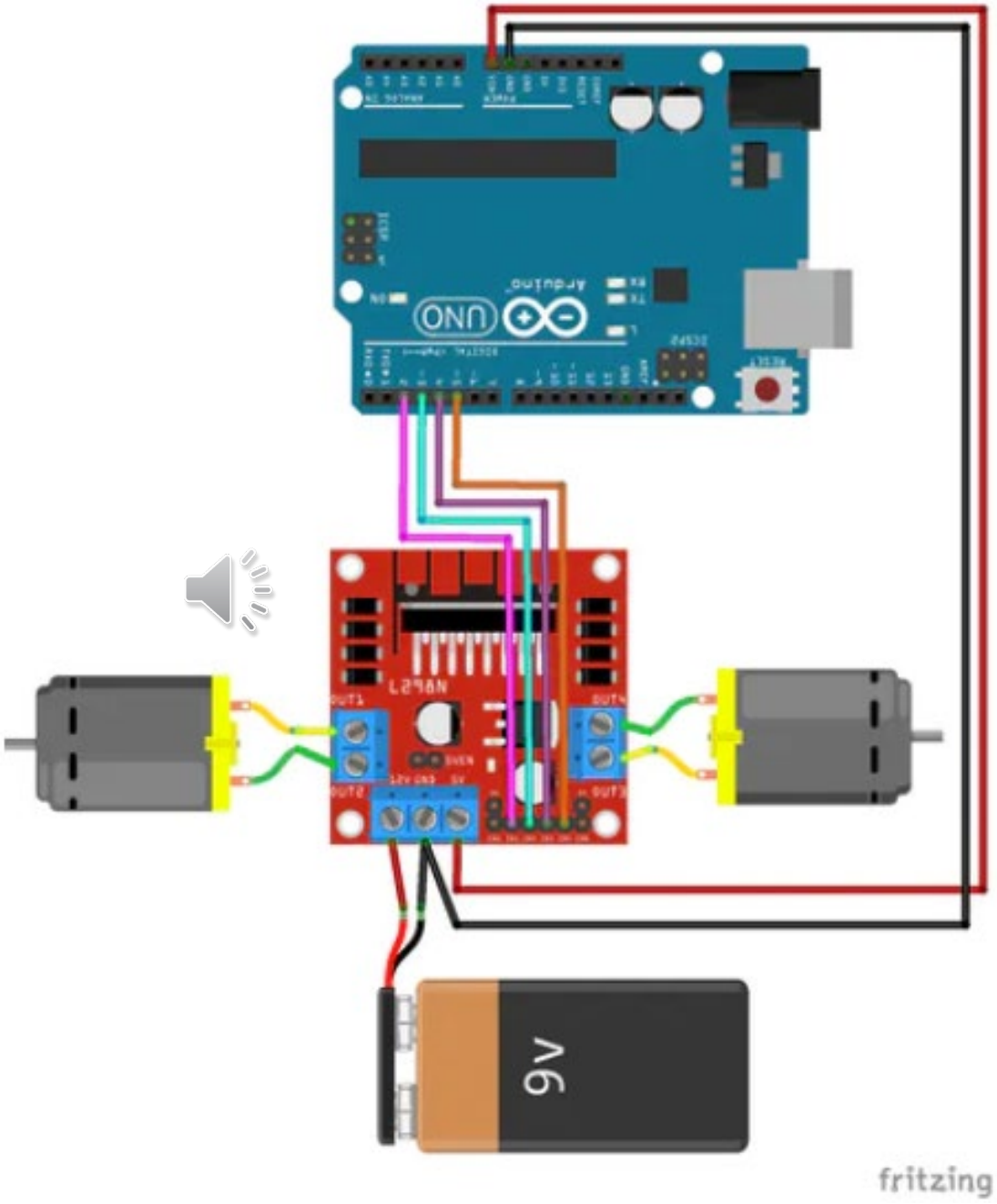
L298 MOTOR DRIVER BOARD

- The L298N Motor Driver is a controller that uses an H-Bridge to easily control the direction and speed of up to 2 DC motors. It operates in a similar way to the L293 that we used previously, but it has a built in voltage regulator and screw terminals that make its use for controlling the robot easier.
- The Arduino web site has an excellent tutorial on using the L298 https://create.arduino.cc/projecthub/ryanchan/how-to-use-the-l298n-motor-driver-b124c5

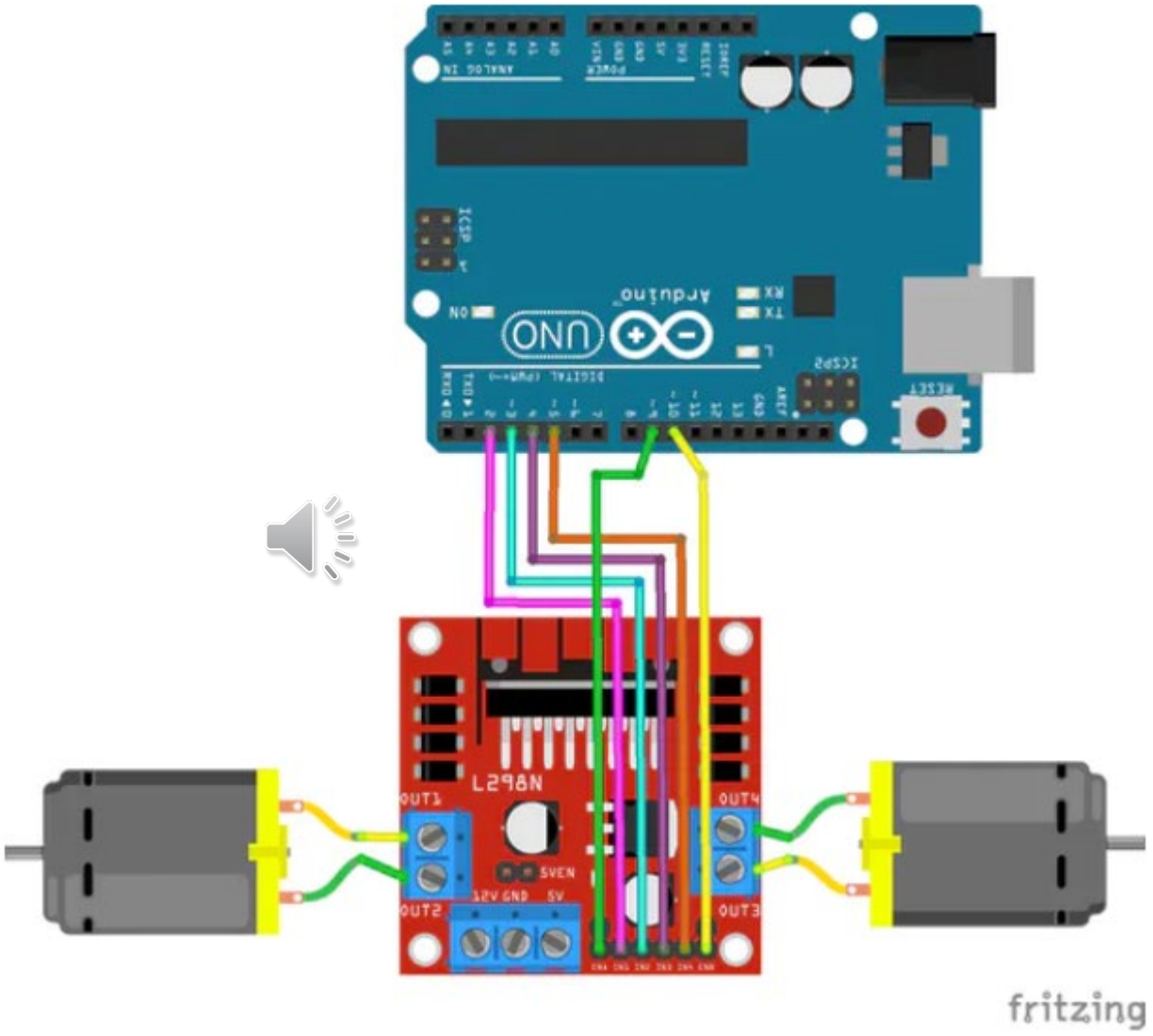


L298 WIRING

- This shows how the L298 should be wired up
- Ensure the ground is connected between the Arduino and the controller



SPEED CONTROL CONNECTIONS



```
/*
Driving of two Bi-directional Motors
*/
/* initial declarations */
int enablePin1 = 9;
int enablePin2 = 10;
int in1Pin = 2;
int in2Pin = 3;
int in3Pin = 4;
int in4Pin = 5;
/* variables for speed of motors */
int leftServoSpeed = 155; // adjust these values between 0 and 255
int rightServoSpeed = 155; // to run at different speeds
void setup()
{
  pinMode(in1Pin, OUTPUT);
  pinMode(in2Pin, OUTPUT);
  pinMode(in3Pin, OUTPUT);
  pinMode(in4Pin, OUTPUT);
  pinMode(enablePin1, OUTPUT);
  pinMode(enablePin2, OUTPUT);
}
void loop()
{
 boolean reverse = 0; //change this to change direction of motors
  setMotor(leftServoSpeed, rightServoSpeed, reverse);
}
void setMotor(int leftServoSpeed, int rightServoSpeed, boolean reverse)
{
  analogWrite(enablePin1, rightServoSpeed); //pwm the enable pin with the speed value for the right motor
  digitalWrite(in1Pin, ! reverse);
  digitalWrite(in2Pin, reverse);
  analogWrite(enablePin2, leftServoSpeed); //pwm the enable pin with the speed value for the left motor
  digitalWrite(in3Pin, ! reverse);
  digitalWrite(in4Pin, reverse);
}
```

TASKS TO BE COMPLETED

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